The Ant Stack

Conceptual Architecture for Embodied Cognitive Systems

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1 Abstract

We propose the Ant Stack: a compact, modular architecture for studying embodied agents through three conceptual layers—`AntBody` (morphology, actuation, sensors), `AntBrain` (functional abstractions inspired by insect circuits), and `AntMind` (active-inference and stigmergic decision-making). Explicit interfaces and sparse-processing assumptions organize the design; they do not establish a complete ant emulation, measured low-power implementation, or validated species preset library.

The proposed methods connect physical and sensory representations to neural-processing and policy-selection models. Navigation, trail following, task allocation, and adversarial-robustness tasks are evaluation protocols, not completed empirical results in this manuscript. Timing, neuron counts, and horizon ranges in the sections below are declared engineering targets or assumptions, not biological measurements or demonstrated real-time capabilities.

The contribution is an architecture and a set of testable interface/model proposals. The companion complexity–energetics workflow supplies analytical counter-based estimates under illustrative coefficients; it does not establish calibration, species validity, or biological efficiency. Independent data, implemented end-to-end task evaluation, and device measurements are required before claiming replication, accuracy, resilience, transfer, or energy superiority.

For myrmecology, this separation offers a way to formulate hypotheses about sensorimotor and collective organization while keeping literature observations distinct from model assumptions. For synthetic systems, swarm robotics and cognitive security are proposed applications, not demonstrated deployment or alignment guarantees.

1.1 Keywords

Myrmecology; ant colony; stigmergy; active inference; Antennal Lobes; Mushroom Bodies; Central Complex; sparse processing; embodied AI; swarm robotics; cognitive security.

2 The Ant Stack

The Ant Stack is a proposed compact, modular architecture for studying embodied agents through body, brain, and mind layers. It emphasizes explicit interfaces and insect-inspired abstractions. Biological plausibility, low-power operation, robustness, and transfer to hardware remain hypotheses to evaluate, not demonstrated properties of a complete ant emulation.

The framework is composed of three primary layers: `AntBody`, `AntBrain`, and `AntMind`. This structure allows for a separation of concerns, where physical simulation, neural architecture, and cognitive modeling can be developed and studied independently while remaining interoperable. The goal is to build upon foundational research in other insects, such as *Drosophila* and *Apis*, to accelerate the development of a sophisticated ant model with wide-ranging applications.

2.1 Proposed Research Directions

- Evaluate `AntBody` I/O, `AntBrain` AL–MB–CX abstractions, and `AntMind` generative-model/stigmergy choices with explicit task metrics.
- Establish source-backed species parameter sets and independently reproducible protocols before claiming species fidelity or one-click empirical replication.

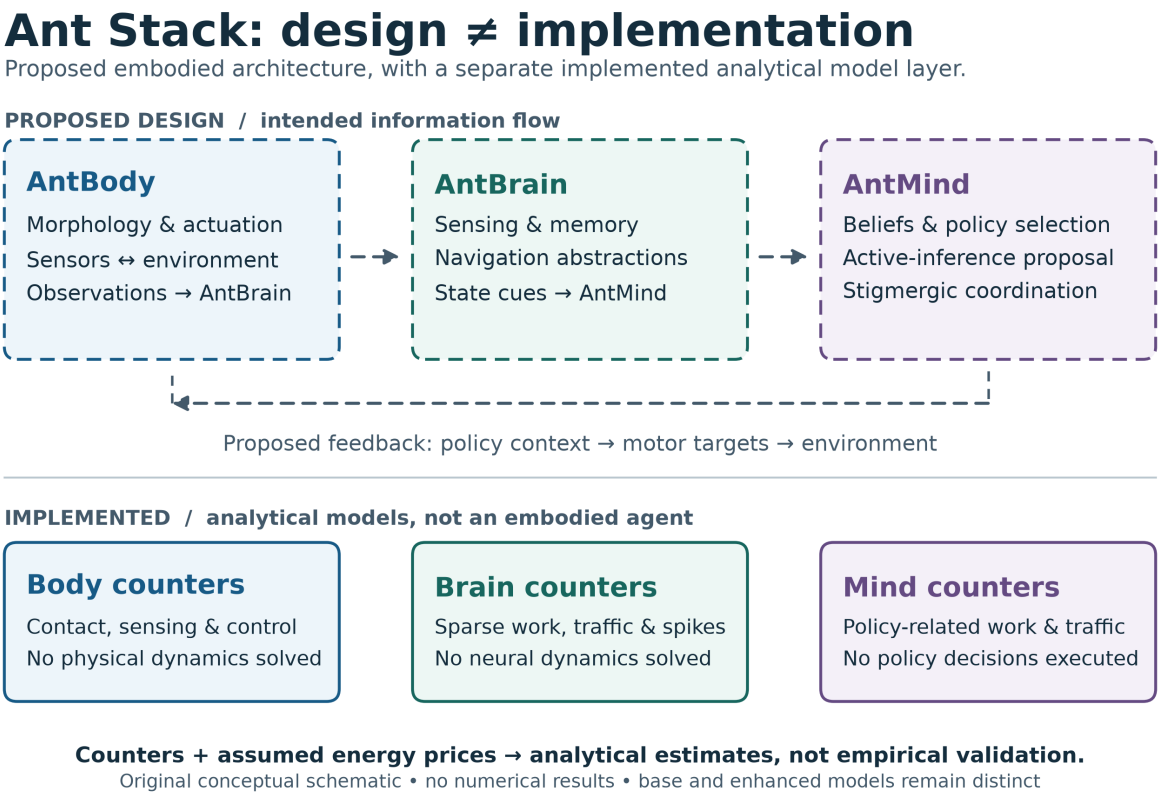


Figure 1: Proposed Architecture and Implemented Analytical Boundary: Original conceptual schematic distinguishing proposed information flow from the implemented workload-count models. It contains no measured data or numerical results; the source and output identities are recorded in the graphical abstract provenance receipt.

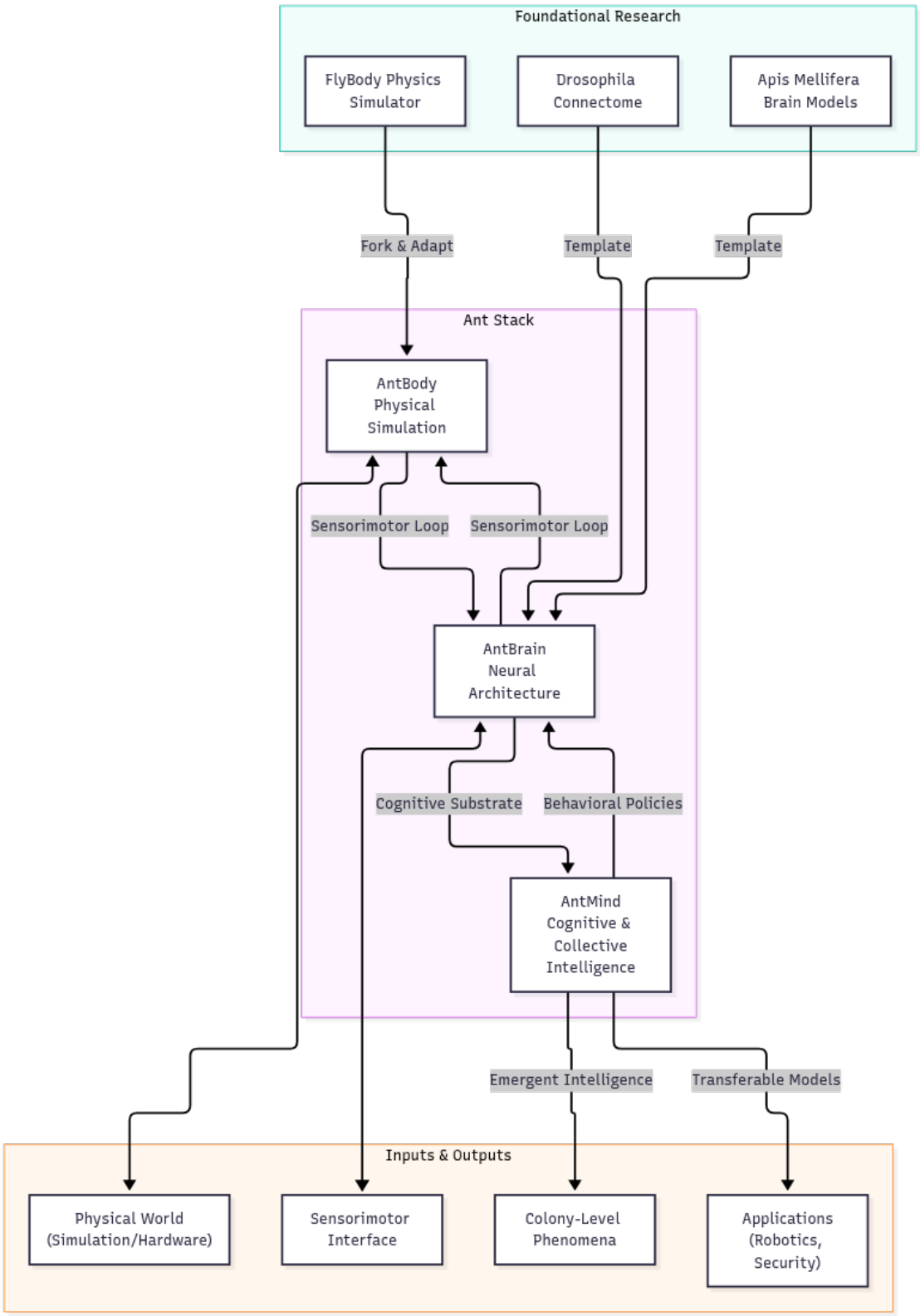


Figure 2: Proposed Body, Brain and Mind interfaces; not a complete implemented agent

- Contributions welcome via pull requests. Keep edits small; include seeds, units, and benchmarks. Propose task variants aligned with the evaluation suite.

2.2 Conceptual Contribution and Evaluation Boundary

The conceptual decomposition supplies a way to formulate testable models of sensorimotor and collective behavior. The companion CE workflow analyzes assumed workload counters, not a completed end-to-end neural/physical ant simulation. All numerical design targets require independent timing, task, and energy evaluation before deployment or scientific validation claims.

3 Component Research and Limits of the Architecture Proposal

This section motivates a proposed architecture using component research in computational neuroscience, robotics, and entomology. Functional fidelity, compact operation, hardware transfer, and cross-species validity are evaluation goals; the present package does not demonstrate them.

3.1 Component Research Motivating the Architecture

- **Neural simulators and cognitive models (e.g., Nengo, Spaun):** Examples of neural/cognitive modeling, not validation of an ant brain. Spaun's reported tasks concern numeral recognition, memory, and motor output (Eliasmith et al., 2012).
- **Neurokernel:** GPU-accelerated, modular simulation of the fly brain; a blueprint for ant-specific simulation
- **Ant-inspired robotics/navigation:** Swarms and navigation (visual odometry, polarized light) validate robust ant-like strategies
- **Field data pipelines:** Increasing availability of arena/field datasets enables data-driven parameterization and validation
- **Tracking/pose tools:** Open-source video tracking and pose estimation (e.g., DeepLabCut, idtracker.ai) enable parameter extraction and validation

3.2 Limits of the Present Proposal

This manuscript does not establish a high-fidelity, closed-loop ant brain/body emulation. The following are limitations of the present proposal, not exhaustive claims that no external research exists:

- **Anatomical evidence:** No complete ant connectome is supplied as this model's input; fly/bee templates require explicit cross-species assumptions.
- **Functional fidelity:** Synapse-level and behavioral fidelity have not been demonstrated here.
- **Real-time implementation:** No embedded full-brain timing benchmark is reported here.
- **Cross-species transfer:** Species-specific calibration and held-out transfer evaluation are required.

3.3 Technical Challenges

- **Sensorimotor integration:** Pheromones, polarized light, chemical gradients, and motor coupling
- **Connectomics:** Incomplete wiring data

- **Hardware miniaturization:** Ant-scale embedded implementation needs device-specific engineering and evaluation; this proposal does not demonstrate it.
- **Data scarcity:** Sparse, noisy, or heterogeneous ecological data complicate validation and benchmarking
- **Standardization:** Lack of common I/O contracts impedes replication and comparison
- **Systems integration:** Bridging simulation and hardware via standardized message schemas (e.g., ROS 2) and unit registries

3.4 AI Concepts Informing the Design

- **Agentic AI and world models:** `AntMind`'s generative model serves as a compact world model for goal-directed behavior
- **Compression over memorization:** A conceptual lens for collective behavior; no universal neuron-count claim or demonstrated compression advantage is inferred here.
- **From tokens to cognition:** Embodiment, grounding, and predictive processing over disembodied token prediction
- **Cognitive security:** Security at the cognitive layer (deception-resilient agents) as a design objective
- **Energy-aware design:** Sparse activity and local learning align with on-edge constraints

3.5 Assumptions and Limits

- No complete ant connectome is used here; cross-insect templates are functional abstractions.
- Real-time embedded operation is unvalidated; timing values are proposed design targets.
- Energy/compute constraints modeled as design targets, not hardware-fitted
- Biological realism is subordinated to functional parsimony where trade-offs are explicit and documented

3.5.1 Prior-Work Evidence and Remaining Integration Gaps

- Prior component work motivates this proposal; it does not validate a full embodied Ant Stack agent.
- Data gaps and hardware constraints justify functional abstraction and simulation-first scope
- Explicit assumptions/limits enable reproducibility and focused progress

4 AntBody: Proposed Morphology and Sensorimotor Interfaces

The proposed `AntBody` would couple morphology, biomechanics, actuation, and sensors through explicit interfaces. `FlyBody` and MuJoCo provide design references; this repository does not implement a FlyBody-to-ant adapter, a MuJoCo/PyBullet dynamics solver, or an executed sensorimotor stream. Its implemented Body workload functions estimate analytical operation counts and assumed energy costs. Reproducing ant contact dynamics, gradient sensing, and actuation latencies remains an evaluation objective.

4.1 Morphology and Biomechanics

- **Segmented body:** Head, thorax, gaster with articulated joints.
- **Six articulated legs:** Multiple degrees of freedom (DOF) per leg for tripod gait and uneven terrain.

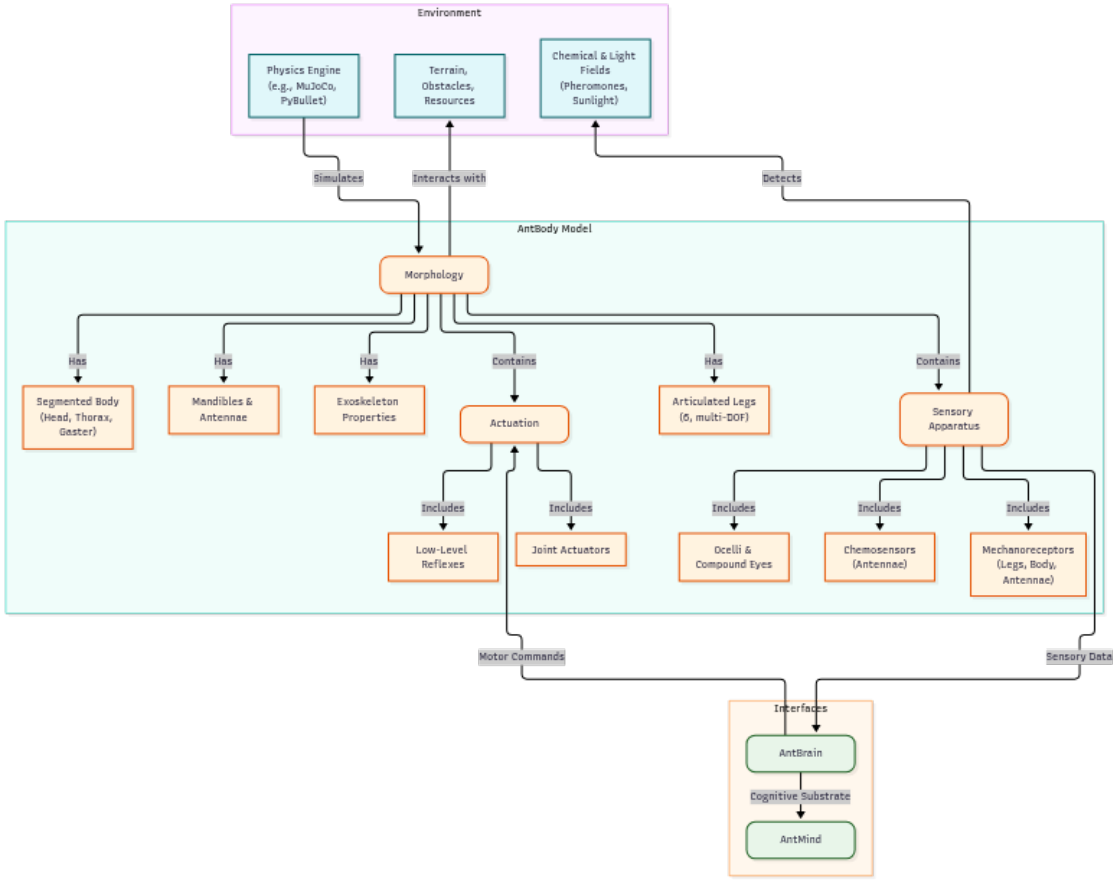


Figure 3: Proposed sensorimotor interfaces; no implemented physics engine is claimed

- **Mandibles and antennae:** Grasping/manipulation; near-field chemosensation.
- **Exoskeleton properties:** Rigidity and mass distribution for realistic contact.
- **Thermoregulation & hydration:** Optional environmental coupling for temperature and humidity tolerance experiments.

4.2 Actuation and Motor Control

- **Joint actuators:** Muscle-like force generation per joint.
- **Low-level control:** Reflexive stabilization (e.g., stance) beneath `AntBrain` commands; optionally PID/impedance at joints for robust terrain contact.

4.3 Sensory Apparatus

- **Chemosensors:** Antennal channels detect pheromone gradients and absolute concentration.
- **Mechanoreceptors:** Leg/body/antennal contact, forces, joint angles/velocities.
- **Ocelli/compound eyes:** Ambient luminance, polarized-light compass, low-res motion/landmarks.
- **Auditory/vibration (optional):** Substrate-borne vibration sensing for alarm/communication experiments.

Goal: a functionally accurate ant nestmate body posing the same control problems a real ant brain evolved to solve.

4.4 Proposed Interfaces (I/O Contract)

All rates, latency/drift budgets, step sizes, and noise ranges below are engineering assumptions or acceptance targets, not measured performance or species-specific data.

- **Observation o_t (**100 Hz**; SI units unless noted):**
  - Chemosensors: K channels/antenna, normalized [0,1]; gradient and absolute concentration
  - Mechanoreceptors: per-leg contact (bool), ground reaction forces (N), joint angles (rad), joint velocities (rad/s)
  - Vision: ocelli luminance (normalized), optional optic flow (px/s), polarized-light compass (deg)
  - IMU (optional): linear acceleration (m/s^2), angular velocity (rad/s)
- **Action a_t (**100 Hz**):**
  - Joint targets: position (rad) or torque ($\mathrm{N} \cdot \mathrm{m}$) per DOF (configurable)
  - Mandible aperture (rad), antennae joint targets (rad)
- **Timing:** Physics step $\Delta t = 1$ ms; control loop acts every 10 steps (100 Hz)
- **Latency budget:** End-to-end sensor→actuator latency target ≤ 20 ms (configurable)
- **Synchronization:** Monotonic timebase; max drift between body and brain clocks ≤ 2 ms; timestamp every observation

Strict units and update rates enable drop-in replacement of bodies and simplify benchmarking across engines.

4.5 Configuration Assumptions and Illustrative Defaults

- Dynamics: position or torque control
- Terrain: flat, rough, slope

- Sensor noise: Gaussian σ per channel
- Pheromone field: on/off, diffusion D, decay λ
- Energy model (optional): per-actuator energy and baseline metabolism for efficiency metrics
- Contact/friction: Coulomb friction coefficients (μ_s, μ_k) and restitution per material
- Proposed sensor/actuator calibration inputs: per-channel offset/gain and instrument logs; no completed auto-calibration campaign is claimed.
- Batch mode: offline rollout/export of o_t, a_t, and internal state for dataset generation

Illustrative assumed defaults: 3–4 DOF per modeled leg, physics step $\Delta t = 1$ ms, 100 Hz control, Gaussian normalized-channel noise $\sigma \in [0.01, 0.05]$. Stability and task suitability require evaluation; these are not physiological measurements.

4.5.1 Sensorimotor Scope and Verification Requirements

- Proposed physics/sensing substrate with explicit I/O targets and units
- Low-level stabilization and robustness require implemented-controller evaluation
- Parameterized terrain, control, noise, and stigmergy for reproducible experiments

4.6 Further Technical Notes and References

- **Dynamics and engines:** MuJoCo and PyBullet are possible simulation tools; a 1 ms step and 100 Hz control are proposed settings, not universal stability guarantees or benchmarks from those tools.
- **Leg kinematics and gaits:** Three or four modeled DOF per leg is an engineering design option, not a calibrated ant morphology.
- **Sensory/navigation context:** Seelig and Jayaraman (2015), Neural dynamics for landmark orientation and angular path integration, studies fly heading circuitry. It is not ant-specific evidence or proof of this controller's timing.
- **Pheromone field:** Diffusion–decay is a proposed environment abstraction; specify space/time units, deposition, boundary conditions and numerical stability before using it to infer biological behavior.

5 AntBrain: Proposed Insect-Inspired Neural Abstractions

`AntBrain` is a proposed functional abstraction inspired by insect neural architecture. Circuit-, cell-, and synapse-level descriptions below specify design choices, not an implemented ant connectome or demonstrated adaptive behavior. The package implements analytical sparse-processing workload counters; independent neural dynamics, learning, and navigation implementations and their validation remain necessary.

The proposed architecture draws inspiration from mapped insect brains, with functional AL/MB/CX abstractions. It does not supply an ant-specific connectome or validate cross-species mappings. Neuron populations, rates, and sparse activity below are configurable engineering assumptions/targets, not measured biological parameter estimates.

5.1 Scope and Assumptions

- Functional, not synapse-accurate; modules align to conserved circuits (AL, MB, CX)
- Proposed 100 Hz closed loop and population range of 10^5–2.5×10^5 modeled neurons are engineering targets; no timing, energy, or species fidelity is demonstrated by those choices

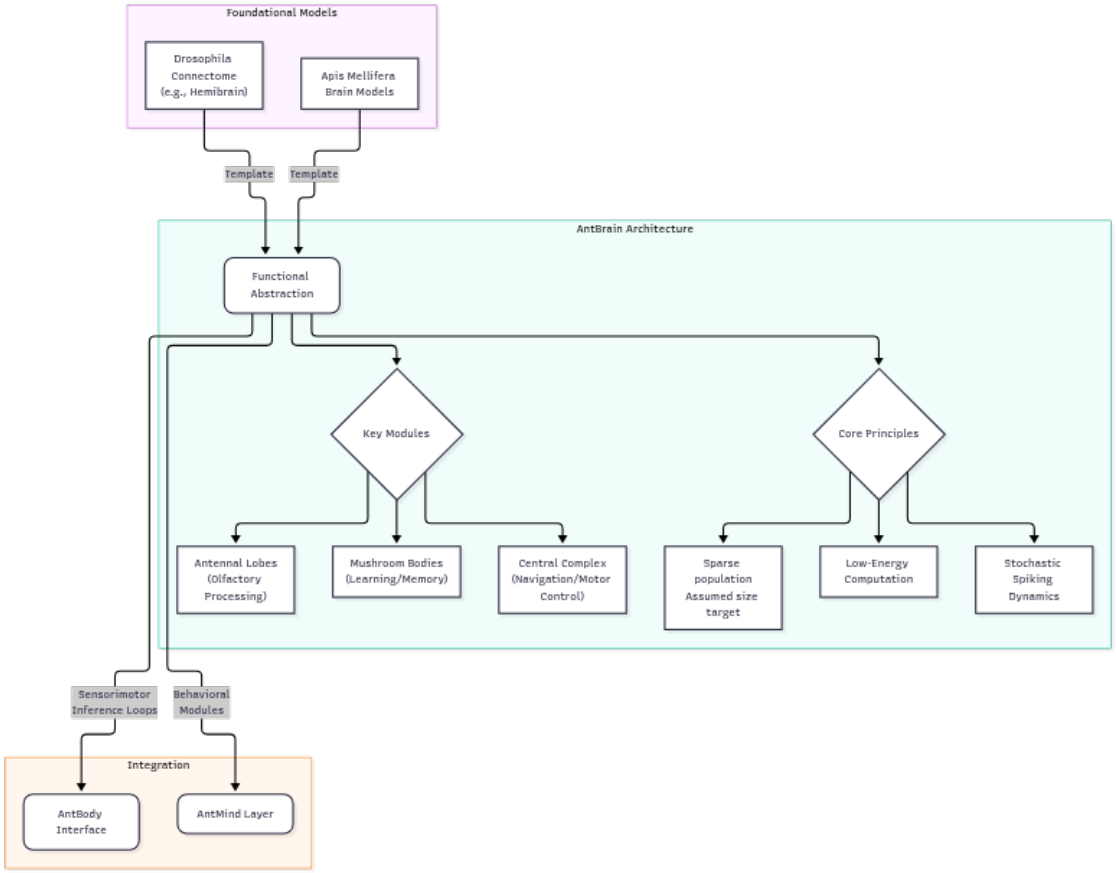


Figure 4: Proposed insect-inspired processing stages and unvalidated neural interfaces

- Local learning (e.g., STDP) with simple modulatory signals; no global backprop
- Noise is a feature: stochasticity aids exploration and regularization; parameters expose variance at module boundaries

5.2 Template Resource: Virtual Fly Brain (VFB)

Virtual Fly Brain (About) is an interactive resource for exploring the neuroanatomy, neuron connectivity, and gene expression of Drosophila melanogaster. It integrates curated literature and image datasets onto a common brain template, enabling cross-search, similarity queries, and 3D comparison of neurons and regions.

VFB can guide region nomenclature and source discovery. Any transfer of its fly data to ant parameters needs an explicit biological justification; no validated ant species override is demonstrated here.

5.3 Design Principles

5.3.1 1. Functional Emulation

Simulate functional roles of key structures rather than replicate every neuron. Favor minimal sufficient mechanisms with measurable interfaces.

5.3.2 2. Neural Sparsity and Energy-Efficiency Targets

- **Population size:** About 250,000 is an illustrative design target, not a measured or universal ant neuron count
- **Optimization:** Low-energy computation and sparse representations
- **Dynamics:** Stochastic spiking and local learning (Hebbian, STDP)
- **Neuromodulation:** Reward/aversive signals (dopamine, octopamine, other biogenic amines) gate plasticity and policy bias with per-task schedules exported for reproducibility

5.3.3 3. Integration

- **Sensorimotor loops:** Tight closed loop with `AntBody`
- **Behavioral modules:** Substrate for higher-level programs orchestrated by `AntMind`
- **Multi-modal fusion:** AL/MB olfaction with visual/vestibular cues into CX heading and action channels with a proposed 100 Hz closed-loop target whose stability/latency must be evaluated

5.4 Proposed Implementation Choices and Timing Targets

- Engine: spiking (Brian2 / Nengo / SpikingJelly) or hybrid rate-based
- I/O: map `AntBody` observations to AL→MB→CX; policy head drives CX motor channel
- Proposed interfaces should expose typed values and unit metadata; end-to-end compatibility requires evaluation.
- **Cross-module timing targets:** AL/MB at 100 Hz and CX at 50–100 Hz are proposed rates, not measured throughput.
- Learning:
  - MB: sparse coding (Kenyon cells); local plasticity with reward/punishment gating
  - CX: ring-attractor heading; soft WTA for action selection

- Targets: configured populations of 10^5–2.5×10^5, sparse activity, and nominal 100 Hz closed loop; none is a measured performance result
- **Memory:** short-term eligibility traces; long-term synaptic consolidation checkpoints for reproducibility; model cards documenting learning rules and seeds
- Neuromodulation API: dopamine/octopamine gating channels exposed via a minimal interface for reproducible learning schedules
- Footprint reporting: track neuron counts, parameter sizes, and update rates per module

5.4.1 Technical Pointers and References

- **Antennal Lobe (AL):** Fly early olfactory processing; Wilson (2013), Early Olfactory Processing in Drosophila: Mechanisms and Principles.
- **Mushroom Bodies (MB):** Fly connectivity context; Caron et al. (2013), Random convergence of olfactory inputs in the Drosophila mushroom body.
- **Central Complex (CX):** Fly heading context; Seelig and Jayaraman (2015), Neural dynamics for landmark orientation and angular path integration.
- **Efficient Spiking Simulation:** Brian2 (docs), Nengo (site), SpikingJelly (GitHub).
- **Virtual Fly Brain (VFB):** Structural templates and nomenclature (about).

These are anatomical and simulation resources, not evidence that this architecture meets its proposed 100 Hz loop or energy targets. The companion CE counter model supports only `random` and `structured` patterns, not a calibrated biological topology.

5.5 Key Neural Circuits and Their Functions

While a full connectome is absent, we can model the primary functions of key, conserved insect brain regions.

- **Antennal Lobes (AL):** This is the first-order olfactory processing center. It receives input from chemosensors on the antennae and organizes odor information into a combinatorial code. Different odors evoke unique patterns of glomerular activation, forming a "scent signature" that is passed to higher brain centers.
- **Mushroom Bodies (MB):** The MB is a critical center for associative learning and memory, homologous to the hippocampus in vertebrates. It receives processed olfactory information from the AL and integrates it with other sensory modalities and internal state information. Its sparse coding scheme, enforced by a large number of Kenyon cells, is ideal for forming and storing specific memories, such as linking an odor to a food reward or a threat.
- **Central Complex (CX):** This is a highly-structured region crucial for spatial navigation, goal-directed behavior, and action selection. It integrates sensory cues (especially visual information like polarized light for sky-compass navigation) to maintain a representation of the ant's heading and orientation relative to its environment. It plays a key role in translating high-level goals (e.g., "return to the nest") into specific directional motor commands.

5.5.1 Selected References

- See `Resources.md` for core literature on insect brain organization and active inference, and for datasets/tools (e.g., VFB) used to template modules.

5.5.2 Neural Design Assumptions and Integration Targets

- Compact, biologically grounded AL→MB→CX control stack with sparse coding and local plasticity
- Tight interfaces with `AntBody` (sensorimotor) and `AntMind` (policy/context)
- Efficiency and configurability prioritized over full connectomic fidelity

6 AntMind: Conceptual Active Inference and Stigmergic Coordination

`AntMind` is the proposed cognitive layer linking `AntBrain` abstractions to policy selection and collective coordination. The equations and interfaces specify hypotheses for individual and colony behavior; the package does not execute active-inference decisions or demonstrate colony-level cognition. Whether unchanged local rules scale from one agent to a colony is a testable design hypothesis, not a result established here.

6.1 Scope and Assumptions

- Active Inference links perception and action via a compact generative model
- Symbolic abstraction emerges from grounded sensorimotor predictions
- Colony cognition uses stigmergy and sparse sharing; no centralized controller
- Proposed short horizons (up to 2 s) are design choices, not demonstrated tractability. The CE model's H_p instead denotes decision steps; converting to seconds requires a specified planning-step duration.

6.2 Key Concepts

6.2.1 Individual Cognition: Active Inference

- **Predictive processing:** Perception updates a generative model to reduce prediction error
- **Free Energy Principle:** Actions minimize expected free energy over time
- **Symbolic grounding:** Symbols emerge from sensorimotor predictions (supports the "triple equivalence")
- **Risk sensitivity:** Preferences encode risk/ambiguity attitudes; policy selection trades off exploration vs exploitation

6.2.2 Collective Cognition Hypothesis: Stigmergy

- **Stigmergy:** Environmental traces (pheromones) coordinate behavior; the trail is the memory
- **Distributed cognition:** Memory, decision-making, and learning are shared across agents and environment
- **Footprint target:** Population size is configured in the neural design, not a universal 250,000-neuron fact or evidence of emergent intelligence.
- **Resilience hypothesis:** Environmental memory and local priors may support recovery from deception; adversarial task evaluation is required.

6.2.3 Proposed Symbolic Grounding from Sensorimotor Predictions

The stack offers a pathway from sub-symbolic processing to symbolic reasoning via grounded predictions. Symbols are treated as compressed, re-usable predictions tied to tasks and sensory contexts.

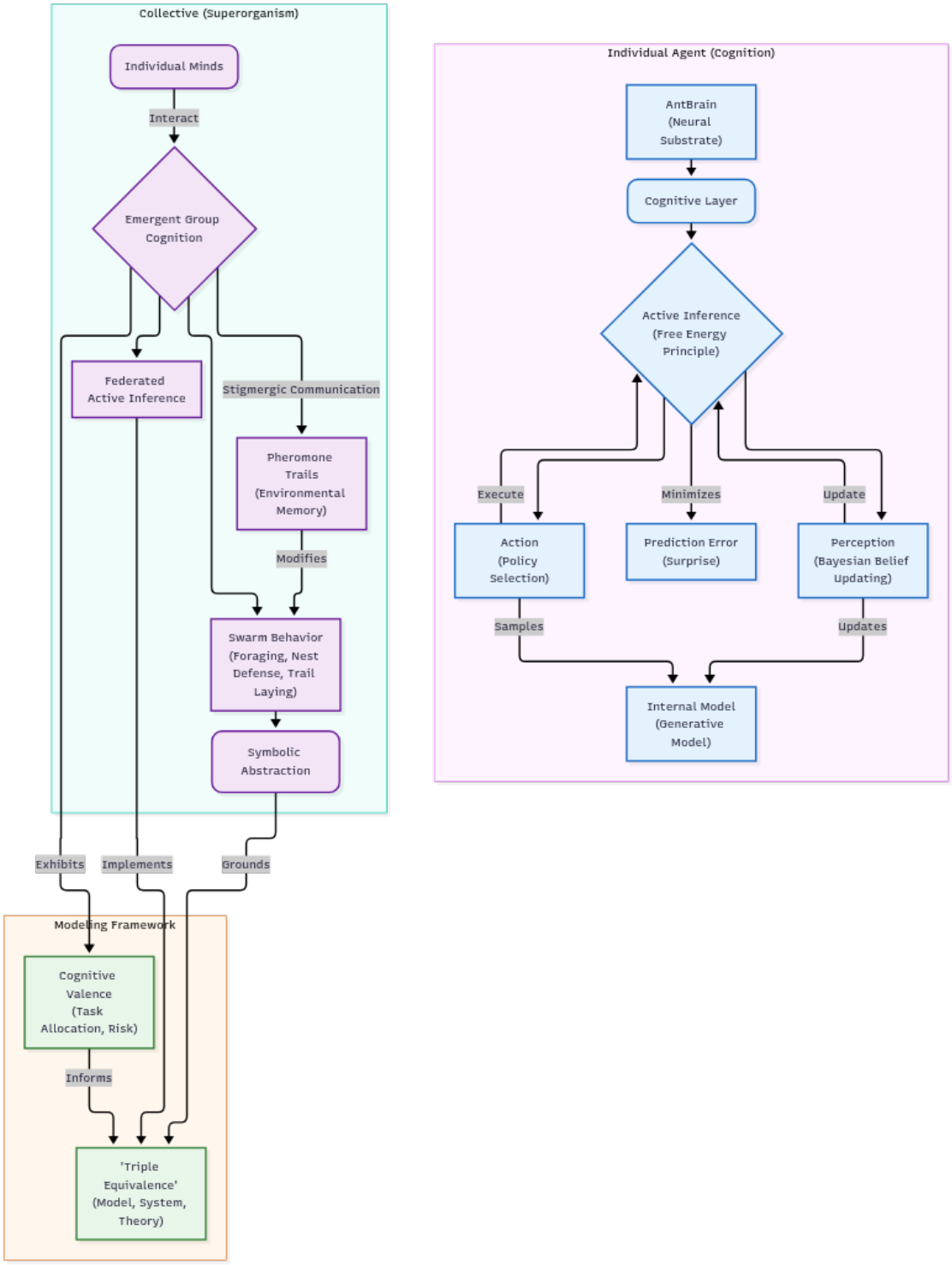


Figure 5: Proposed individual inference and colony coordination; closed-loop behavior is unverified

6.3 Proposed Single-Agent Generative Model

- Latent state s_t: pose, heading, internal drive (hunger), local pheromone expectation
- Observation o_t: from `AntBody` I/O
- Action a_t: joint targets
- Preferences: priors over outcomes (food proximity↑, energy cost↓, collision↓)
- Constraints: energy budget and temperature/humidity limits (optional) shape expected free energy
- Proposed update: variational inference and policy selection over an illustrative 0.5–2.0 s horizon; this is not measured task performance.
- Proposed rates: control at 100 Hz and policy update at 10 Hz, not demonstrated device throughput.
- Diagnostics: report expected free energy terms (risk, ambiguity), action entropy, and policy dwell time
- Diagnostics: report EFE decomposition (epistemic/pragmatic value), action entropy, policy dwell time, and constraint violations

6.4 Proposed Diffusion–Decay Pheromone Field

- A proposed explicit field update is $c_{t+1} = (1 - \lambda\Delta t)c_t + D\Delta t\nabla_h^2 c_t + q_t$, with time step Δt, spatial discrete Laplacian ∇_h^2, and deposits q_t in concentration units.
- Specify D in area/time, λ in inverse time, grid spacing, boundary conditions, nonnegativity, and stability. Reward-linked deposits are a modeling choice, not a measured pheromone law.
- Agents follow ∇c; following probability rises with $|\nabla c|$
- Interfaces: deposit/read operations are unit-aware; decay and diffusion are version-pinned for replication
- Stability: saturation/clipping on c_t and deposit rates to prevent runaway trails; optional anisotropic diffusion under wind

6.4.1 Technical Pointers and References

- **Active inference foundations:** Friston et al. (2015), Active inference and epistemic value.
- **Ant-colony modeling context:** Friedman et al. (2021), Active Inferants: An Active Inference Framework for Ant Colony Behavior, is an in-silico foraging model, not validation of this architecture or a specific ant neural mechanism.
- **Stigmergy and Ant Trails:** Canonical formulations in ant optimization and diffusion-decay fields (Ant colony optimization, Fick's laws of diffusion).

Conceptual symbolic processing does not eliminate physical compute energy. Policy evaluation must be counted on the hardware where it runs. Base CE Mind uses canonical B, H_p and exact tree-node enumeration; enhanced Mind is a separate bounded leaf-policy model (`max_policies=1000`, or `None` for exact enumeration). No universal 15-step timing boundary or quality-preserving sampling result is established.

6.5 Proposed Colony-Level Arbitration

- Federated active inference: local beliefs shared sparsely via stigmergy; no central controller
- Task allocation: soft constraints via internal drives and pheromone-mediated opportunities
- Safety: monitor for deceptive gradients; trigger re-exploration and attenuate deposit under conflict

- Belief sharing: lossy compression of shared statistics (e.g., recent reward rates, local gradient summaries)

6.5.1 Cognitive Design Goals and Unverified Collective Behavior

- Proposed interfaces connect neural processing, policy selection and collective behavior.
- The generative model and stigmergy dynamics require an implemented closed-loop evaluation.
- Compact policies and cross-agent composition remain design goals, not measured results.

7 Proposed Applications and Evaluation Protocols

The following are proposed applications and evaluation protocols, not demonstrated deployments, calibration standards, or evidence of safety/alignment. Ant-inspired efficiency and robustness remain task-specific hypotheses to evaluate.

Design intent: a single, compact agent architecture that ports from high-fidelity simulation to embedded platforms and multi-agent swarms with minimal change.

7.1 Proposed Cross-Domain Evaluation Metrics

- **Efficiency**: energy per reward; compute per action
- **Robustness**: success under sensor/actuator noise and partial failures
- **Scalability**: performance vs number of agents; communication overhead
- **Alignment**: deviation from intended norms under perturbations/adversaries
- **Reproducibility**: standardized configs, seeds, and dataset/version pinning

7.2 Swarm Robotics

Proposed ant-inspired control for physical-platform evaluation; deployment readiness has not been established.

- **Disaster response:** Navigate debris, locate survivors, perform cleanup in hazardous settings
- **Logistics and construction:** Stigmergic coordination for warehousing and build tasks
- **Environmental monitoring:** Distributed mapping/sampling with trail-guided coverage
- **Fieldable constraints:** low compute/energy budget, lossy comms, intermittent GPS/vision

7.2.1 Evaluation (swarm robotics)

- Time-to-food, path efficiency, energy per reward, success under sensor noise
- Robustness to actuator loss and terrain changes; recovery time
- Communication budget (bandwidth/duty cycle); MTTF/MTTR under staged failures

7.3 Networks and Optimization

- **Routing/load balancing:** ACO grounded in realistic diffusion/decay for routing, caching, congestion
- **Distributed scheduling:** Colony-inspired task allocation under constraints; throughput, fairness, resilience
- **Resilient overlays:** Decoy trails and evaporation parameters for adaptive rerouting under attack

- **Cache placement:** Use pheromone analogs for content popularity and decay-based invalidation

7.4 Cognitive Security

Secure systems at the cognitive layer: resilience to deception, spoofing, and manipulation.

- **Secure swarms:** Test whether decentralized control improves resilience under a declared compromise model; decentralization alone does not guarantee security
- **Defending complex systems:** Colony-style distributed detection/response for critical infrastructure
- **Cognitive honeypots:** Deploy deceptive pheromone fields to study adversary behavior and improve defenses
- **Incident triage:** Prioritize actions via expected free energy; quantify trade-offs under uncertainty

7.4.1 Evaluation (cognitive security)

- Adversary success probability vs defense; detection AUC on deceptive signals; recovery time
- False-positive/negative balance; cost-of-delay under competing alerts

7.5 Biosurveillance & Biodefense

- **Cognitive anomaly detection:** Learn baselines from sensors; flag deviations
- **Disease spread modeling:** Reuse pheromone diffusion/decay for pathogen propagation
- **Sentinel networks:** Agent subsets specialized for detection/triage to reduce false alarms
- **Adaptive sampling:** Allocate sensing effort via stigmergy when anomalies persist

7.5.1 Evaluation (biosurveillance)

- Anomaly PR-AUC on simulated outbreaks; detection latency vs false alarms
- Spatial coverage vs resource budget; escalation pathways

7.6 Foundational AI Research

- **Emergent intelligence:** Study complex goal-directed behavior from simple local rules
- **Benchmark proposal:** Evaluate energy-efficient AI designs after device calibration and task-specific validation
- **Embodiment ablations:** Vary body/brain/mind couplings to test contributions to generalization
- **Neural efficiency:** Map performance–energy trade-offs; quantify sparse spiking benefits

7.6.1 Evaluation (foundational AI)

- Navigation success on mazes/rough terrain; memory capacity via odor–reward associations
- Policy entropy, sample efficiency, and transfer across terrains/species presets

7.7 AI Alignment and Safety

- **Alignment hypothesis:** Test when local incentives support or conflict with collective welfare; natural organization does not guarantee alignment

- **Emergent goals:** Test designs that avoid undesirable emergent behaviors without central control
- **Norm compliance:** Measure deviation under resource stress and adversarial signals
- **Value handshakes:** Encode minimal priors that align individual incentives with group welfare

7.7.1 Evaluation (alignment)

- Colony social welfare under scarcity; deviation from norms under adversarial pheromones
- Collateral risk vs objective attainment; robustness to deceptive gradients

7.7.2 Proposed Testbeds and Requirements for Fair Comparisons

- The application domains are proposed testbeds, not demonstrated implementations or performance results.
- Comparisons require implemented baselines, matched measurement boundaries and declared evaluation protocols.

7.7.3 Terminology Note

- We follow community guidance on language and update in consultation with myrmecologists
- Domain experts are invited to propose task variants and metrics; small PRs welcome

8 Research Implications, Limits, and Evaluation Priorities

8.1 Proposed Uses in Myrmecology

- **Empiricism → executable theory**: Trail dynamics, polarized-light navigation, and associative olfaction become testable modules (pheromone field, ring-attractor CX, AL→MB learning), enabling hypothesis tests and cross-species comparisons
- **Individuals → superorganisms**: Stigmergic composition supports division of labor, trail reinforcement, and resilience with field-measurable levers (diffusion, decay, deposit)
- **Experiment ↔ field loop**: Parameters map to measurable field quantities, enabling calibration and out-of-sample prediction
- **Data leverage**: Bibliometrics and curated datasets guide species/trait prioritization and parameter ranges

8.2 Ecological Questions for Future Evaluation

- **Urgency-tuned trails**: Context-dependent deposition/following to study exploration–exploitation shifts
- **Environmental stability**: Species-parameterized noise, energy costs, terrain for robustness under heat/drought/fragmentation
- **Ant–plant mutualisms**: Resource–reward dynamics to test protection-for-nectar feedbacks and colony outcomes
- **Collective risk management**: Study alarm signaling, quorum thresholds, and evacuation under threat
- **Navigation cues**: Integrate polarized light and landmarks into CX to test cue-combination strategies

8.3 Limitations

- No ant connectome is supplied as input here; fly/bee abstractions require justified cross-species mapping, not a claim that no external ant connectomic work exists
- Real-time embedded operation and proposed task outcomes are not validated by this conceptual manuscript or by the companion analytical counter sweeps
- Field validation requires collaboration and standardized data collection protocols
- Chemical ecology is complex; microclimate and substrate effects can shift pheromone dynamics and sensing

8.4 Future Directions

- **Species libraries**: Parameter presets for representative taxa/ecoregions
- **Learning mechanisms**: Local plasticity beyond STDP (neuromodulated Hebbian) under energy budgets
- **Collective tasks**: Nest construction, brood care, adversarial decoys to test colony cognition
- **Validation**: Benchmarks vs field/arena datasets (trail formation, polarized-light homing, task allocation)
- **Open protocols**: Share standardized tasks and seeds to enable cross-lab replication
- **Tooling**: Programmatic APIs for species presets, parameter sweeps, and experiment manifests
- **Systems bridges**: Minimal ROS 2 bindings and message schemas; unit-registry enforcement end-to-end
- **Security**: Counter-deception benchmarks and ablations (e.g., spoofed gradients, adversarial deposits)

9 Foundational Research and Resources

These are conceptual inputs and attributed literature findings, not validation of this architecture or calibration of its analytical coefficients. Project websites are resources, not independent evidence of implemented integrations.

9.1 Foundational Projects

- FORMINDEX: bibliographic exploration that can guide source discovery; parameter choices still require primary evidence.
- MetaInformAnt: a data-integration resource, not evidence of a completed Ant Stack ingestion/calibration workflow.
- Active Inferants, Friedman et al. (2021): *Active Inferants: An Active Inference Framework for Ant Colony Behavior*, an in-silico colony-foraging model. Its published proof of principle is not empirical validation of the proposed Body–Brain–Mind architecture or any specific ant neurocognitive mechanism.
- Virtual Fly Brain, Court et al. (2023): *Virtual Fly Brain—An interactive atlas of the Drosophila nervous system.* It supports anatomical data discovery and nomenclature; transferring fly anatomy to an ant requires a justified cross-species mapping.

9.2 Neural and Cognitive Context

- Wilson (2013), Early Olfactory Processing in Drosophila: Mechanisms and Principles: a review of early fly olfactory processing, not an ant-specific input-channel estimate.

- Caron et al. (2013), Random convergence of olfactory inputs in the Drosophila mushroom body: fly mushroom-body connectivity context, not a calibrated topology in the CE counter model.
- Seelig and Jayaraman (2015), Neural dynamics for landmark orientation and angular path integration: fly heading-circuit context, not proof that an Ant Stack controller runs at the proposed update rate.
- Friston et al. (2015), Active inference and epistemic value: decision-theoretic foundations; expected free energy is not automatically physical energy in joules.

9.3 Ecological Literature: Attributed Findings

- **Abundance and biomass:** Schultheiss et al. (2022), The abundance, biomass, and distribution of ants on Earth, estimate approximately 20 quadrillion individuals and about 12 megatons of dry carbon, with substantial extrapolation and geographic/data gaps. Their dry-carbon biomass comparison exceeds wild birds and mammals combined; it is not a comparison of counts or wet body mass. These are literature estimates, not outputs of this repository. The paper reports more than 15,700 named species **and subspecies** in its contemporary taxonomy, not a current verified count of species alone.
- **Invasion impacts:** Tercel et al. (2023), Non-native ants drive dramatic declines in animal community diversity: A meta-analysis, report taxon-level local abundance and richness reductions of 42.79% and 53.56% in their selected invaded versus uninvaded, relatively undisturbed sites. These are heterogeneous meta-analytic estimates, not universal effects for every species or habitat and not Ant Stack simulations.
- **Ecosystem services and disservices:** Anjos et al. (2022), The effects of ants on pest control: a meta-analysis, find context-dependent pest-control effects, including benefits for non-honeydew-producing pests and disservices involving honeydew producers/natural enemies. Shaded-crop results do not imply uniformly beneficial ants.
- **Bird–ant interactions:** Avilés (2024), The evolutionary ecology of bird–ant interactions: a pervasive but under-studied connection, reviews recorded interactions involving 1122 bird species and 47 ant genera. This is the scope of the review, not a complete ecological census or a modeled interaction network here.
- **Elevational diversity:** Szewczyk and McCain (2016), A Systematic Review of Global Drivers of Ant Elevational Diversity, report mid-elevation peaks, low plateaus, and monotonic declines, with contingent multi-driver explanations rather than one universal climate law.
- **Biogeography:** Wang et al. (2024), Global biogeographic regions for ants have complex relationships with those for plants and tetrapods, provides a distributional/phylogenetic regionalization. It is ecological context, not a source of neural or energy coefficients.

9.4 Proposed Data and Tool Integration

Pose/tracking tools such as DeepLabCut and idtracker.ai, anatomical resources, and ecological databases can help define datasets and hypotheses. Unit handling (Pint) and messaging (ROS 2) are possible interface tools, not claims of implemented bindings. Imported data require raw-data provenance, licenses, units, species/caste context, and independent validation before supporting manuscript claims.

Primary identities and literature scopes above were checked against publisher/DOI records or the original full texts. The former Wilson, Caron, and Seelig links were mismatched and have been replaced with their checked intended records; no unrelated DOI is used as evidence for an insect

circuit. No species-specific neural population, metabolic rate, CoT, calibration accuracy, or real-time capability is inferred from this resource list.

10 Appendices

10.1 A. Proposed Reproducibility Checklist

This is an acceptance protocol, not a record of completed trials, instrument calibration, species validation, or implemented infrastructure.

- Fixed random seeds (report all)
- Explicit I/O rates and units (see `AntBody`)
- Engine versions: physics, spiking/NN libraries, OS
- Environment configs: terrain, pheromone D, λ, deposit rules
- Agent configs: neuron counts, learning rules, energy budgets
- Evaluation protocols and metrics; trials; confidence intervals
- Data/code availability; exact commit SHA
- Registered experiment manifests (YAML/JSON) to enable one-click reruns
- Model cards: describe objectives, assumptions, limits, and validation datasets
- Unit tests: per-module interface and invariants; scenario tests for end-to-end loops
- Unit registry and versions (e.g., Pint) recorded; message schema versions (e.g., ROS 2) pinned
- Continuous integration manifest for experiment reruns and document/PDF builds

10.2 B. Proposed Species Parameterization Protocol

No preset is validated by naming a species. Bind each numerical parameter to primary data for species/caste, colony, environment, and units; otherwise mark it as an illustrative assumption. Evaluate held-out individuals/colonies separately from fitting data.

- `AntBody`: leg DOF, segment masses, antenna channel count K, sensor noise σ, gait presets
- Pheromone: diffusion D, decay λ, baseline deposit, reward/urgency modulation
- `AntBrain`: AL glomeruli count $\leftrightarrow$ K, MB sparsity (Kenyon cell ratio), CX ring size
- `AntMind`: preference priors, policy horizon, update rates; colony size, sharing frequency
- Colony ecology: resource density, nest locations, predator risk profiles (for security experiments)
- Provenance: cite sources for parameter choices and note uncertainties
- Energy/compute envelope; actuator latency distributions; friction coefficients per terrain/material

10.3 C. Proposed Task-Evaluation Templates

- Navigation: maze/slope/rough terrain; success, path ratio, energy per reward
- Trail following: deposit reward-linked trail; formation time, stability, traffic
- Task allocation: mixed stochastic arrivals; utilization, latency, resilience
- Adversarial robustness: deceptive pheromones; deviation from norms, recovery time
- Security drills: spoofed gradients, sensor jamming; detection latency and false-positive rate
- Reporting: include seed ranges, CIs, ablations, and failure cases with traces
- Specify communication-overhead budgets and failure-injection schedules

11 Glossary

- **Active Inference (AIF)**: Agents minimize variational free energy through belief updating and policy selection
- **AL (Antennal Lobes)**: First-order olfactory processing; glomerular odor code
- **Alignment (AI)**: Consistency of behavior with intended objectives under perturbation
- **CX (Central Complex)**: Hub for heading, orientation, and action selection; often a ring attractor
- **Energy budget**: Compute/energy cost per unit behavior; critical for embedded intelligence
- **Eligibility trace**: Short-term memory of recent activity used to gate plasticity updates
- **Generative model**: Predicts sensory inputs; supports policy selection via expected free energy
- **Kenyon cells**: Sparse-coding neurons in Mushroom Bodies; associative learning
- **MB (Mushroom Bodies)**: Learning and memory; multimodal integration
- **Pheromone field**: Diffusion–decay concentration for stigmergic communication
- **Ring attractor**: Network motif for continuous heading representations
- **Sparse spiking**: Low average activity for energy efficiency
- **Stigmergy**: Indirect coordination via environmental modification (e.g., pheromone trails)
- **Tripod gait**: Hexapod locomotion using alternating tripods
- **Expected free energy (EFE)**: Decision-theoretic objective involving risk, ambiguity and epistemic value; not automatically thermodynamic energy in joules
- **Variational free energy (VFE)**: Variational bound used for inference; physical energy requires a separately justified model and units
- **STDP (Spike-Timing-Dependent Plasticity)**: Local synaptic update rule driven by pre-/post-spike timing
- **WTA (Winner-Take-All)**: Competitive selection mechanism used for action selection or sparsification
- **Quorum sensing**: Thresholded group decision mechanism relevant to task allocation and evacuation
- **I/O contract**: Explicit interface specifying message contents, units, and update rates between modules
- **Unit registry**: Programmatic system for tracking and enforcing physical units across computations
- **Epistemic value**: Information gain component of EFE
- **Pragmatic value**: Goal-directed (extrinsic) utility component of EFE

11.1 Abbreviations

- **AIF**: Active Inference
- **AL**: Antennal Lobes
- **MB**: Mushroom Bodies
- **CX**: Central Complex
- **DOF**: Degrees of Freedom
- **IMU**: Inertial Measurement Unit
- **PR-AUC**: Precision–Recall Area Under Curve